

Lecture 16, Mar 04 2026.

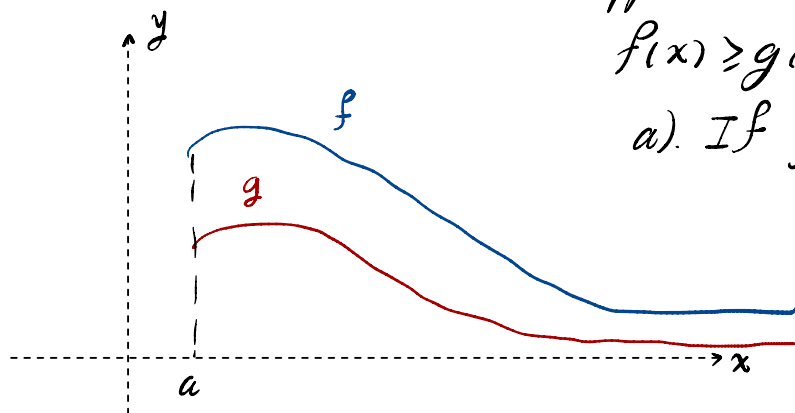
* A comparison test for improper integrals.

Thm

Suppose that f and g are continuous with $f(x) \geq g(x)$ for $x \geq a$.

a). If $\int_a^\infty f(x) dx$ is convergent then $\int_a^\infty g(x) dx$ is convergent.

b). If $\int_a^\infty g(x) dx$ is divergent, then $\int_a^\infty f(x) dx$ is divergent.



Example. Show that $\int_0^\infty e^{-x^2} dx$ is convergent.

Sln. let consider $f(x) = e^{-x}$ We claim that $f(x) \geq g(x) = e^{-x^2}$.

$$e^{-x} = \frac{1}{e^x} \geq e^{-x^2} = \frac{1}{e^{x^2}}$$

then
$$\int_0^\infty e^{-x} dx \geq \int_0^\infty e^{-x^2} dx.$$

Now
$$\int_0^\infty e^{-x} dx = \lim_{t \rightarrow \infty} \int_0^t e^{-x} dx = \lim_{t \rightarrow \infty} \left\{ -e^{-x} \right\}_0^t$$
$$= \lim_{t \rightarrow \infty} \{ 1 - e^{-t} \} = 1.$$

Since $\int_0^\infty e^{-x} dx$ is convergent then by comparison thm

$$\int_0^\infty e^{-x^2} dx \text{ is convergent.}$$

Example. Evaluate $\int_1^\infty \frac{1+e^{-x}}{x} dx$.

Sln. We claim that $\int_1^\infty \frac{1+e^{-x}}{x} dx$ is divergent.

let consider $f(x) = \frac{1}{x}$, $g(x) = \frac{1+e^{-x}}{x}$ and notice that

$$\frac{1}{x} \leq \frac{1+e^{-x}}{x} \text{ for all } x \geq 1.$$

$$\text{So, } \int_1^{\infty} \frac{1}{x} dx \leq \int_1^{\infty} \frac{1+e^{-x}}{x} dx.$$

$$\begin{aligned} \int_1^{\infty} \frac{1}{x} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx \\ &= \lim_{t \rightarrow \infty} \ln |x| \Big|_1^t \\ &= \lim_{t \rightarrow \infty} \{ \ln t - \ln 1 \} \\ &= \lim_{t \rightarrow \infty} \ln(t) = \infty. \end{aligned}$$

By the comparison thm $\int_1^{\infty} \frac{1+e^{-x}}{x} dx$ is divergent.

10.1 Curves Defined by Parametric equations.